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Experiment 1: Assembly and commissioning of a telescope

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1 Introduction

Telescopes are known to be the main instrument that astronomers use to study the cosmos. They can be divided on different clasifications, depending on the range of light which they observe. Observing in wavelengths visibles by human eye, we could find optical telescopes. The first refracting telescope used to observe the sky was constructed by Galileo. Its simplified configuration consists on one biconcave and one biconvex lense. Another refracting configuration can be formed adding a Barlow Lens in a optical system. It consists on a optical tube used to increase the effective focal lenght (EFL), which leads on a increase on the magnification of the telescope.

Before using a telescope, it is necesary to characterize some important quantities that are useful to have a reference of what type of objects can be seen using our instrument. The ratio of the transverse dimentions of the final image formed by any optical system, as telescopes, to the corresponding dimension of the object is called transverse magnification [2]:

$$M = \frac{h_i}{h_o} = \frac{d_i}{d_o} = \frac{\theta_i}{\theta_o} \tag{1}$$

Where h_i and h_o means the height of the image and the object respectively, d_i and d_o are the distance from the lens to the image and the object respectively [2]; θ_i and θ_o are the angles in which the image is seen through the telescope and the angle under which the object is seen by the unaided eye [3]. Also we can determinate the magnifying power using the diameter of the entrance d_{ent} and exit pupils d_{ex} of the telescope as:

$$M = \frac{d_{ent}}{d_{ex}} \tag{2}$$

We can also measure the field of view (FOV) in a simple experiment, placing an object distant to the telescope and registering the area seen by it. Then we can use trigonometry to obtain the relation:

$$FOV = tan^{-1}\left(\frac{h_o}{d_o}\right) \tag{3}$$

From them we obtain the result in degrees or radians. Then we can pass to seconds of arc using the known relation:

$$1rad = 206265" \tag{4}$$

Finally, it is interesting to know the angular resolution of the telescope. The common way to obtain it is using the wavelenght of the observation and the diameter of the aperture, as $R = \lambda/D$. But without using filters, we are observing in all the range visible by human eye. A simple way to obtain it is to write two parallel lines in a paper in a distance d_{lines}, and measure the distance from the telescope

to the paper d_p in which the lines are not distinguishables one from the other. Here we can use trigononometry to say:

$$R = tan^{-1}\left(\frac{d_{lines}}{d_p}\right) \tag{5}$$

We can measure the error of a division as approximately the sum of the relative errors:

$$\frac{\delta_q}{q} \approx \frac{\delta_x}{y} + \frac{\delta_y}{y} \tag{6}$$

In this work, we assembled a refracing telescope known as *Galileoscope* [cita pagina], with the objective to characterize an optical instrument. We measured the magnification and FOV for three different configurations avaliables and also the angular size to one of them. Also, we learned how to handle lenses and make simple observations. In Section 2 is shown the procedure followed to assemble the telescope and take the measurements. In Section 3 are the results obtained for the parameters of the telescope. Finally, in Section 4 is presented a discussion and ending remarks about the experiment.

2 Experimental Procedure

There was assembled the *Galileoscope* following the instructions avaliable in the guidelines [1]. We had three configurations of the *Galileoscope*. The expected magnifications on each one was 17x, 25x and 50x; so we refer to each telescope as them. Some important parameters are also reported in Table 1.

The final assembling of the telescope in the laboratory looks as seen:

2.1 Magnification

For this experiment we used:

- A galileoscope.

Parameter	Value
Objective diameter	50 mm (two inches)
Objective focal length	500 mm (1/10)
Eyepiece focal length	20 mm
Magnification	25x (50x with Barlow)
Field of view	16 mm (22mm with Barlow)
Eyepiece eye relief	16mm (22mm with Barlow)
Eyepiece barrel diameter	1 +1/4 inches (31 +3/4 mm)

Table 1: Optical characteristics of *Galileoscope*, obtained from the guidelines of the class.



Figure 1: Telescope assembled in the lab.

- Measuring tape (graded in mm).
- Ruler of 30 cm and 90 cm (graded in mm).

We started with the experiment inside the instrumentation laboratory. There we placed a ruler in the wall. Then, we placed our telescopes on one of the tripods avaliable (there was three for five telescopes, but only two of them had stability to put the telescope) in turns and observed by them using the Galilean 17x configuration. We noted that was needed the maximum distance possible to see a little portion of the ruler. By the telescope, I observe 25 ± 1 mm of the ruler. Then we measured the distance from the ruler to the eyepiece using a measuring tape, in my case I obtained 680 ± 10 mm. Then I also measure the size of the entire ruler from the distance of the eyepiece using

my fingers on the measuring tape. I observed the ruler of 300 mm as 23 ± 1 mm size.

For the next configurations we decided to go outside the lab, to perform the measurements. Due to the few tripods avaliable we decided to perform all the measurements in one time. Here, a classmate was placed as a test subject to observe by the telescope. There was another person in the side, to measure the height from him observed by the entire exit pupil of the telescope, using a ruler. There was placed the measuring tape on the floor measuring a distance of 29± 0.001 m from the telescope to the classmate. In the telescopes side there was a person observing at the 25x configuration and another at 50x. There is a person in the middle of both to pass the messages from one side to the another. When the person was found by one telescope, the observer was indicating the range visible from the pupil. In the other side, there was measured this range. For the 25x there was observed 556 ± 1 mm. For 50x there was observed 231 ± 1 mm. Both measurements was made by the same distance between the person and the telescopes. We also estimated the distance in which the eye is placed to observe to the pupil, using glasses we estimated that value near to 2 cm.

2.2 Field of View

For this experiment we used the same materials and procedure described in 2.1.

2.3 Angular resolution

For this experiment we used the same materials as 2.1, adding:

- A board marker.
- Graph paper.

We drew two parallel lines in the graph paper, separated by 1 mm. Then a person was placed in front of the 25x configuration until the observer found him. There was needed more distance to not distinguish one line from another, so we marked the point where was located the 29 m of distance between the person and the telescope. From there, the person was walking away until the lines were not distinguished from the eyepiece. There was used the measuring tape from this point to measure the distance. We obtained a value of 58.55 ± 0.001 m.

Finally, we used the ruler to measure the diameter to the eyepiece for each configuration. For 17x, there was measured 10 ± 1 mm. For 25x and 50x, which uses the same, there was measured 12 ± 1 mm.

3 Results and Analysis

For each telescope there were measured the magnification and FOV. For the 17X, we calculated the magnification measuring the ratio of the angles where the image is seen by the telescope and the telescope is seen by the eye, using the last case described in (1):

$$M_{17x} = \frac{tan^{-1}\left(\frac{0.3m}{0.68m}\right)}{tan^{-1}\left(\frac{0.023m}{0.68m}\right)} = 12.289 \qquad (7)$$

Then we calculated the error as the error of the division:

$$\frac{\delta_M}{M} = \frac{\delta_{\theta_i}}{\theta_i} + \frac{\delta_{\theta_o}}{\theta_o} \qquad (8)$$

Where the error of each angle can be obtained as $\delta_{\theta_o} = \delta_{FOV_{17x}} = 9.5 \cdot 10^{-5}$ rad (calculated later) and $\delta_{\theta_i} = 0.003$ rad. Putting them on (8), we obtain a result of 12.289 ± 1.54.

For the 25x and 50x configuration we also measured the angles in which the image is visible by the telescope and the object is seen by eye:

$$M_{25x} = \frac{tan^{-1}\left(\frac{0.012m}{0.025m}\right)}{tan^{-1}\left(\frac{0.555m}{29m}\right)} = 23.387 \qquad (9)$$

Including the error, we had $M_{25x} = 23.387 \pm 0.75$. Doing the same, we obtained $M_{50x} = 56.18 \pm 1.28$

For the FOV, we used equation (3) and (4) to obtain:

$$\begin{aligned} FOV_{17x} &= tan^{-1}\left(\frac{0.025m}{0.68m}\right) \\ &= 0.367rad \cdot 206265 \\ &\approx 7579.86" \end{aligned} \tag{10}$$

$$\begin{aligned} FOV_{25x} &= tan^{-1}\left(\frac{0.555m}{29m}\right) \\ &= 0.019rad \cdot 206265 \\ &\approx 3947.004" \end{aligned} \tag{11}$$

Following the same to the 50x, we obtained $FOV_{25x} \approx 1642.97"$.

For each quantity we measured its associated error, using (7). For 17x we have:

$$\begin{aligned} \frac{\delta_{FOV_{17x}}}{FOV_{17x}} &= tan^{-1}\left(\frac{\delta_{d_o}}{d_o} + \frac{\delta_{h_o}}{h_o}\right) \\ &= tan^{-1}\left(\frac{0.1mm}{25mm} + \frac{10mm}{6800mm}\right) \\ &= 0.005rad \end{aligned} \tag{12}$$

Then,

$$\begin{aligned} \delta_{FOV_{17x}} &= (0.005rad \cdot 0.019rad) \cdot 206265 \\ &\approx 378.49" \end{aligned} \tag{13}$$

So we had a FOV_{17x} of 7579.86" $\pm$ 378.49". Following the same to obtain the errors, we have FOV_{25x} of 3947.004" $\pm$ 15.68" and $FOV_{50x} \approx 1642.97" \pm 3.05"$.

Also, there was chosen the 25x configuration to measure the angular resolution. Using the data obtained from 2.3 and equation (5) with (3):

$$R = tan^{-1}\left(\frac{10^{-3}m}{58.55m}\right) = 3.523'' \tag{14}$$

With the relative error obtained by the uncertainty divided by the value of the quantities:

$$\frac{\delta_{d_{lines}}}{d_{lines}} + \frac{\delta_{d_o}}{d_o} = \frac{10^{-4}m}{10^{-3}m} + \frac{10^{-2}m}{58.55m} = 0.1002 \tag{15}$$

Where we estimated $\delta_{d_{lines}}$ and δ_{d_o} from the grading of the instruments and the distance from the visual observation of the meauring tape. Then we put that on equation (5) and multiplied by the true value to obtain the uncertain:

$$\frac{\delta_R}{R} = tan^{-1}\left(\frac{\delta_{d_{lines}}}{d_{lines}} + \frac{\delta_{d_o}}{d_o}\right) = 0.9998rad \tag{16}$$

Finally:

$$\delta_R = 0.9998rad \cdot R = 1.705 \cdot 10^{-6} \cdot 206265 \approx 0.35'' \tag{17}$$

We reported a value of the angular reslution of 3.25 " $\pm$ 0.35 ".

4 Discussion and Remarks

One of the most important thing to take into account in this experiment was the errors of the procedure. We can divide them into sistematics and random. One sistematic error could be the distance measured from the telescope and the object (or person). Especially on the outside measurements, this distance was not exactly measurement from the eyepiece of the telescope, because we placed the measuring tape on the floor, for stability. Another sistematic error was the use of the fingers and the arms, to remark the longitude visible by the telescope. This marks could be add more uncertainties on the measurement of this quantity (h_o). A random error which is always present is the minimal unit that could measure our instruments. In this experiment we used rulers and tapes graded in mm. Using a sophisticated instrument like a meter foot could be more exact. Finally, the most known way to

Parameter	17x	25x	50x
Magnification	12.289 ± 1.54	23.387 ± 0.7	56.18 ± 1.28
FOV	7579.86" ± 378.49"	73947.004" ± 15.68"	1642.97" ± 3.05"
Resolution		3.25 " ± 0.35 "	

Table 2: Summary of the parameters obtained for each telescope configuration.

minimize the errors are take more measurements and take the mean of them. By the theorem of central limit, the mean value would be approaching to the true value of the quantity measured.

To summarize, in this experiment we learned how to assemble and use a refracting telescope. We also proposed a method to measure the magnification, FOV and angular resolution of different configurations. We remark the importance to perform these analysis to characterize each telescope. In this way, we will know which celestial objects could be observables by our instrument.

References

[1] galileoscope.org. *Galileoscope Assembly Instructions*. IAU, 2009.

[2] Eugene Hecht. *Optics*. Pearson Education India, 2012.

[3] Benjamin King Johnson. *Optics and optical instruments: an introduction with special reference to practical applications*. Courier Corporation, 1960.